8. OCEANWATCH - Decentralized Illegal Fishing Detection

This mission employs a Synthetic Aperture Radar (SAR) satellite constellation to identify illegal fishing vessels within protected marine zones. By utilizing blockchain for transparent enforcement, the system enables NGOs, fishing communities, and governments to coordinate responses through smart contracts, overcoming the high costs and lack of transparency found in traditional monitoring systems.

Satellite Requirements

- **Orbit:** Sun-synchronous LEO 580 km, 6-satellite constellation for 6-hour global revisit
- **Coverage:** Priority zones - Exclusive Economic Zones (EEZs), marine protected areas, Arctic/Antarctic waters
- **Payload:**
    - SAR radar (X-band, 5m resolution, day/night all-weather imaging)
    - AIS receiver (Automatic Identification System - tracks ships broadcasting position)
    - Secondary: Optical camera (10m resolution for visual confirmation)
- **Power:** 120W peak (SAR is power-hungry), 50W average
- **Mass:** <200 kg per satellite (includes large SAR antenna)
- **Comms:** X-band 150 Mbps downlink (SAR imagery is data-intensive)

Web3 System

Data Flow: Satellite SAR detects vessel in protected zone → Cross-references with AIS database (is ship broadcasting legal ID?) → If dark ship (no AIS) or in restricted zone, uploads SAR image + GPS to IPFS → Smart contract records violation with timestamp → NGOs/coast guards alerted automatically → Fishing community members can verify (local knowledge) → Verified violations trigger: coast guard dispatch, fishing license suspension, community rewards for reporting → All enforcement actions recorded on-chain (transparent accountability)

Key Smart Contracts:

1. **VesselRegistry** - Links ship IDs (IMO numbers) to blockchain addresses, tracks fishing licenses
2. **ProtectedZoneDefinition** - Geographic boundaries of no-fishing zones stored on-chain
3. **ViolationDetection** - Automated detection when SAR+GPS shows vessel in restricted area
4. **CommunityVerification** - Local fishing communities confirm/dispute violations (earn tokens for accuracy)
5. **EnforcementAction** - Records coast guard responses, license suspensions, fines paid
6. **RewardDistribution** - Splits fines between: enforcement costs, community reporters, ocean conservation fund

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. Technical Constraints & Requirements

1. Advanced Detection: Eliminating "Dark Vessels"

- **SAR Payload (Synthetic Aperture Radar):** Traditional optical satellites are often rendered useless by cloud cover or nighttime. Our X-band SAR provides **all-weather, 24/7 imaging**. It can detect the metallic hull of a ship through thick storms with **5m resolution**.
- **AIS/SAR Cross-Referencing:** Every legal ship must broadcast an AIS (Automatic Identification System) signal. Our system performs a real-time comparison: if the Radar sees a ship but there is no AIS signal, it is flagged as a **"Dark Vessel"** likely engaged in illegal activity.

- **Global Revisit Rate:** A 6-satellite constellation in **Sun-synchronous LEO (580 km)** ensures we revisit any point on Earth every 6 hours. This eliminates the "blind spots" that illegal fishers currently exploit.

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2. Decentralized Evidence & Verification

- **IPFS Evidence Storage:** Once a violation is detected, the SAR image and GPS coordinates are hashed and uploaded to **IPFS (InterPlanetary File System)**. This ensures the visual evidence of the crime is decentralized and cannot be deleted or modified by corrupt officials.
- **Community Verification:** To add a layer of human intelligence, local fishing communities can verify or dispute the data using their local knowledge. This **"wisdom of the crowd"** increases the accuracy of the system and earns community members tokens for their participation.

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3. Transparent Enforcement via Smart Contracts

- **Automated Violation Detection:** A smart contract pre-loaded with **ProtectedZoneDefinitions** (geographic boundaries) instantly logs a violation the moment a ship crosses a digital fence.
- **Unbiased Enforcement:** The **EnforcementAction** contract records every response—from Coast Guard dispatches to fines paid. Because this is on-chain, it is publicly visible, preventing cases from being "swept under the rug."
- **Economic Incentives:** The **RewardDistribution** contract automatically splits collected fines between the Coast Guard, the local community reporters, and ocean conservation funds, creating a self-sustaining financial loop.

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4. Technical & Engineering Profile

- **Power Management:** Operating SAR is energy-intensive (**120W peak**). The <200 kg satellite design includes high-efficiency solar arrays and large battery reserves to support active radar pulses.
- **Data Throughput:** SAR imagery is high-volume. The system utilizes an **X-band 150 Mbps downlink** to ensure that detection-to-alert latency is kept at a minimum, allowing for rapid interception.
- **Secondary Verification:** An onboard **10m Optical Camera** provides visual confirmation during daylight hours to support the primary radar data, creating a multi-layered legal case against violators.